

Usability Testing Report: Software Engineering

Team #16, The Chill Guys

Hamzah Rawasia

Sarim Zia

Swesan Pathmanathan

Burhanuddin Kharodawala

April 4, 2026

1 Revision History

Date	Developer	Notes
March 29th	Team #16, The Chill Guys	Initial scaffold: sections, methodology shell, placeholders for post-study results.

2 Symbols, Abbreviations and Acronyms

symbol	description
HCI	Human–Computer Interaction
SUS	System Usability Scale
CSUQ	Computer System Usability Questionnaire (optional instrument)
UT	Usability test; task identifiers UT-1 , UT-2 , ...
FR	Functional Requirement (from SRS)
NFR	Nonfunctional Requirement (from SRS)
SRS	Software Requirements Specification
API	Application Programming Interface
T	Task identifier in session scripts (e.g., T1, T2)

Contents

1	Revision History	i
2	Symbols, Abbreviations and Acronyms	ii
3	Introduction	1
3.1	Purpose and Scope	1
3.2	Relation to Requirements	1
3.3	Document Structure	1
4	Goals of Usability Testing	2
5	Methodology	2
5.1	Study Design	2
5.2	Participants	3
5.3	Environment and Materials	3
5.4	Session Outline	3
5.5	Data Recording	4
6	Task Scenarios	4
6.1	Scenario UT-1: First-time access and verification	4
6.2	Scenario UT-2: Rider—find or request a ride	4
6.3	Scenario UT-3: Driver—offer a ride	5
6.4	Scenario UT-4: Matching or trip follow-up	5
6.5	Scenario UT-5: Profile or settings (optional)	5
7	Session Questions and Survey Instruments	5
7.1	Pre-session Background Questions	5
7.2	Post-task Probing Questions	6
7.3	Post-session Interview Questions	6
7.4	Standardized Questionnaires	7
8	Metrics and Quantification	7
8.1	Task Performance	7
8.2	Subjective Measures	7
8.3	Analysis	8
8.4	Thresholds (optional targets)	8

9	Results	8
9.1	Participant Summary	8
9.2	Quantitative Summary	8
9.3	Qualitative Observations	9
9.4	Screenshots or Figures	9
10	Discussion and Recommendations	9
10.1	Interpretation	9
10.2	Recommendations	9
10.3	Traceability	9

List of Tables

1	Task scenario summary (fill task times and assist counts after sessions).	6
---	---	---

List of Figures

This document describes a systematic usability evaluation plan and (after sessions are run) quantitative and qualitative results for Hitchly, a campus-focused rideshare application for the McMaster community. Task identifiers **UT- n** denote usability scenarios in Section 6. Post-session sections (Results and Discussion) are completed after data collection.

3 Introduction

3.1 Purpose and Scope

This report documents a **systematic, rigorous, and quantified** usability evaluation of Hitchly, aligned with HCI practice. It specifies the **goals** of testing, the **tasks** participants perform, the **questions** and **survey instruments** used to capture subjective feedback, and the **metrics** used to analyze performance and satisfaction.

The evaluation targets the mobile experience (and any supporting flows such as email verification) for McMaster-affiliated riders and drivers. It complements verification activities in the Verification and Validation (VnV) plan by focusing on learnability, efficiency, error tolerance, and satisfaction for representative users rather than on code-level correctness alone.

3.2 Relation to Requirements

Usability objectives align with non-functional expectations in the SRS for understandable, low-friction flows (e.g., sign-up, offering or requesting a ride, confirming a match, payment-related steps where applicable). Findings may be traced to specific FR/NFR identifiers when reporting issues or recommendations.

3.3 Document Structure

Section 4 states test goals. Section 5 describes participants, setting, and procedure. Section 6 lists task scenarios. Section 7 covers moderator questions and surveys. Section 8 defines quantitative measures and analysis. Sections 9 and 10 are populated after sessions. Appendix 10.3 holds full questionnaire text.

4 Goals of Usability Testing

The following goals guide session design and success criteria. They should be measurable using the metrics in Section 8.

1. **Onboarding and trust:** Assess whether participants can complete registration, McMaster-affiliated verification, and role selection (rider/driver) without undue confusion or abandonment.
2. **Core rider journey:** Evaluate discoverability and completion of finding or requesting a ride, understanding trip details, and confirming participation within the app.
3. **Core driver journey:** Evaluate offering a ride, setting or understanding route and schedule-related inputs, and managing passenger-related actions as implemented.
4. **Matching and trip state:** Observe comprehension of match suggestions, trip status, and navigation between related screens.
5. **Error recovery:** Record where users deviate, hesitate, or commit errors, and whether they recover without moderator intervention (within a reasonable threshold).
6. **Subjective satisfaction:** Quantify perceived ease of use and confidence via standardized scales (e.g., SUS) and structured debrief responses.

Team members may refine goal wording after piloting the script with one internal participant to ensure tasks remain feasible on the build under test.

5 Methodology

5.1 Study Design

Usability testing uses **moderated task-based sessions**. Each participant works through a fixed set of scenarios (Section 6) on a prepared device or the participant’s device, in a consistent build (e.g., Expo development client or release candidate). The moderator explains the purpose, obtains consent, and encourages a **think-aloud** protocol unless a participant finds it distracting (note the exception in session notes).

5.2 Participants

- **Inclusion:** McMaster students, staff, or faculty (or individuals who match the intended domain, e.g., test accounts with @mcmaster.ca) who are willing to sign consent and have not developed the feature under test.
- **Sample size:** Target a minimum number of sessions (e.g., 5–8) consistent with course guidance; record actual N in Section 9.
- **Roles:** Mix riders and drivers where possible, or assign persona per session and document role in the participant log.

5.3 Environment and Materials

- **Device:** iOS and/or Android; screen recording if permitted by consent.
- **Network:** Stable connection; document if testing against staging or local API.
- **Test data:** Pre-seeded accounts or anonymized personas; reset state between sessions as needed.
- **Ethics:** Brief verbal or written consent covering recording, data handling, and voluntary withdrawal. Store notes and recordings per course or institutional policy.

5.4 Session Outline

1. Welcome and consent (5 min).
2. Background questionnaire (Appendix 10.3, optional short form).
3. Task scenarios in a fixed order for comparability, or counterbalanced order across participants (document order).
4. Post-task and post-session questions (Section 7).
5. SUS or equivalent survey (Appendix 10.3).
6. Debrief and thanks.

5.5 Data Recording

For each task, record: start/end time, success or failure, number and type of assists (minimal prompt vs. direct help), and brief qualitative notes (quotes, confusion points). Use a session log template (spreadsheet or form) referenced in Section 9.

6 Task Scenarios

Each scenario below maps to goals in Section 4. Update wording to match the exact labels and navigation in the build under test. Success criteria should be **binary** or scored with explicit rules for partial credit if used.

6.1 Scenario UT-1: First-time access and verification

- **Context:** You are a new McMaster student who has just installed Hitchly.
- **Task:** Create an account with a McMaster-affiliated email, complete any verification step required, and reach the main home or dashboard screen.
- **Success criteria:** Account created; email verification completed (or flow correctly attempted per test account setup); lands on post-auth home without moderator rescue beyond one standard hint.

6.2 Scenario UT-2: Rider—find or request a ride

- **Context:** You need to get to campus for a class on a typical weekday.
- **Task:** Use the app to search for or request a ride matching your need (origin, destination, time window as prompted by the moderator).
- **Success criteria:** Participant submits or selects a ride request (or equivalent primary action) and can state what happens next; no critical misunderstanding of cost or commitment if shown.

6.3 Scenario UT-3: Driver—offer a ride

- **Context:** You drive to campus and want to share seats.
- **Task:** Create or publish a ride offer with the details the moderator specifies (route endpoints, time, seats).
- **Success criteria:** Ride offer visible to participant as “posted” or equivalent; required fields completed without persistent validation errors.

6.4 Scenario UT-4: Matching or trip follow-up

- **Context:** A compatible ride or match is available (moderator may set state in advance).
- **Task:** Accept, decline, or confirm the match/trip as appropriate to the prototype state.
- **Success criteria:** Participant completes the intended state transition or explains blockage clearly; note any UI ambiguity.

6.5 Scenario UT-5: Profile or settings (optional)

- **Task:** Update a non-sensitive profile field or locate notification/payment settings.
- **Success criteria:** Participant finds the correct area within a time threshold (set in Section 8) or documents failure.

7 Session Questions and Survey Instruments

7.1 Pre-session Background Questions

Q1: How often do you commute to or from McMaster (or campus)?

Q2: Do you usually drive, ride as a passenger, or use transit?

Q3: Have you used any rideshare or carpool apps before? Which?

Q4: What device are you using today?

Table 1: Task scenario summary (fill task times and assist counts after sessions).

ID	Focus	Notes
UT-1	Onboarding / verification	
UT-2	Rider journey	
UT-3	Driver journey	
UT-4	Match / trip state	
UT-5	Profile / settings	

7.2 Post-task Probing Questions

After each scenario (Section 6), ask as appropriate:

- PT1:** What did you think that screen was asking you to do?
- PT2:** Was anything surprising or unclear?
- PT3:** On a scale of 1 (very difficult) to 7 (very easy), how easy was that task? (Record numeric response.)
- PT4:** Did you feel you could trust the information the app showed (e.g., cost, other user, timing)?

7.3 Post-session Interview Questions

- PS1:** Overall, how would you describe your experience using Hitchly today?
- PS2:** What would you change first if you could?
- PS3:** Would you use this for real commuting if it were available? Why or why not?
- PS4:** Any concerns about safety, privacy, or payments?

7.4 Standardized Questionnaires

- **System Usability Scale (SUS):** Administer the standard 10-item SUS immediately after tasks; compute the standard 0–100 score per participant. Full wording appears in Appendix [10.3](#).
- **Optional supplementary Likert blocks:** e.g., trust in campus verification, clarity of pricing—use 5- or 7-point scales with anchors “Strongly disagree” / “Strongly agree”; include items in Appendix [10.3](#).

All numeric survey responses feed the quantitative analysis described in Section [8](#).

8 Metrics and Quantification

This section defines how usability is measured to satisfy **rigorous, quantified** reporting. Fill aggregate values in Section [9](#) after data collection.

8.1 Task Performance

- **Task success rate:** For each UT-*n*, proportion of participants who meet success criteria without failure. Report per-task and overall.
- **Time on task:** Elapsed time from moderator prompt to success or abandonment; report median, interquartile range, and outliers.
- **Assists:** Count moderator prompts (minimal hint vs. substantive help); define weighting if a composite “error score” is used.
- **Critical incidents:** Count usability problems by severity (blocks task / major / minor); use a simple rubric agreed before sessions.

8.2 Subjective Measures

- **SUS score:** Per participant and mean $\pm$ standard deviation across *N*.
- **Post-task ease (1–7):** Mean per task from PT3 (Section [7](#)).
- **Optional Likert composites:** Mean scores per construct (e.g., trust, clarity).

8.3 Analysis

- Summarize quantitative results in tables; use figures (e.g., bar chart of success rates) if required by the course.
- Thematic analysis on open-ended responses: tag recurring themes and link to screenshots or timestamps when recordings exist.
- Map top issues to SRS FR/NFR or design modules where applicable for traceability.

8.4 Thresholds (optional targets)

Document any targets set before testing (e.g., at least 80% success on UT-1, median SUS ≥ 68). Revise after pilot if unrealistic for the current build.

9 Results

This section is completed after usability sessions are conducted.

9.1 Participant Summary

- Number of participants N : *(record after sessions)*.
- Role mix (riders / drivers / both): *(record after sessions)*.
- Build or version tested: *(record after sessions)*.

9.2 Quantitative Summary

Insert tables for:

- Task success rates per UT- n .
- Time on task (median, IQR) per task.
- Assist counts or weighted error scores.
- Mean SUS score $\pm$ SD; distribution if helpful.
- Post-task ease ratings (mean per task).

9.3 Qualitative Observations

Summarize recurring themes, representative quotes (anonymized), and critical incidents with severity counts.

9.4 Screenshots or Figures

Include annotated screenshots or stills from recordings where they illustrate findings (with participant consent as applicable).

10 Discussion and Recommendations

This section is completed after results are analyzed.

10.1 Interpretation

Relate findings to the goals in Section 4. Discuss trade-offs (e.g., verification friction vs. trust). Note limitations (sample size, prototype fidelity, test environment).

10.2 Recommendations

Prioritize actionable UI/UX or copy changes. Use a table with columns: **Issue**, **Evidence** (task ID, quote, metric), **Severity**, **Proposed change**, **Owner** / **status**.

10.3 Traceability

Link high-priority recommendations to SRS requirements or GitHub issues as appropriate for the capstone project.

Appendix — Survey Instruments and Consent

Consent Script (Outline)

- Purpose of the session; voluntary participation; right to stop.
- Recording (screen/audio) if used; storage and who may view data.
- No penalty for withdrawal; approximate duration.
- Contact for questions.

System Usability Scale (SUS)

For each item, use a 5-point scale from **Strongly disagree** (1) to **Strongly agree** (5). Score using the standard SUS formula (odd items: score = response -1 ; even items: score = $5 - \text{response}$; sum $\times 2.5$ for 0–100).

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.
4. I think that I would need the support of a technical person to be able to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Optional Likert Items (Trust and Clarity)

Use 7-point Likert scales (1 = Strongly disagree, 7 = Strongly agree). Adapt wording after pilot if needed.

1. I trust that other users on this app are affiliated with McMaster as claimed.
2. The app's pricing or cost information was clear.
3. I understood what would happen after I confirmed a ride or match.
4. I felt my personal information was handled appropriately for a student project prototype.

Session Log Template (Fields)

- Participant ID (anonymous); date; moderator; device/OS; build.
- Per task: start time, end time, success Y/N, assists, notes.
- SUS total; optional Likert means.